Digital Signal Processing Exercise Sheet 1

The ECU Sees Only Signals

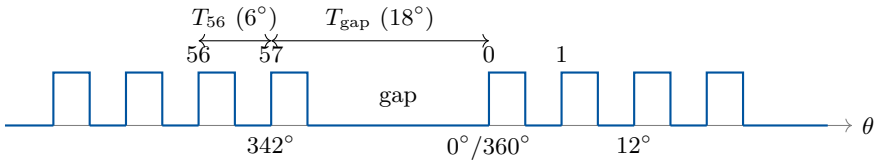
Script: Chapter 1 (The ECU Sees Only Signals)

Learning goals

- Classify the engine sensor signals (continuous/discrete time, analog/binary amplitude, deterministic/stochastic, periodic/aperiodic) and know their characteristic time scales.
- *Bonus (optional):* describe sensor signals quantitatively: mean, RMS, extreme values, autocorrelation as a numerical test for periodicity.
- Extract a first piece of information from a raw signal: crank angle and engine speed from the 60-2 tooth timing, including the effect of the timer resolution.
- Synchronise crank and cam signal to determine the 720° phase of the engine cycle (cylinder identification).

Background: the 60-2 trigger wheel

The crank wheel of the virtual engine has 60 tooth positions (6° each), two of which are missing. The Hall sensor delivers a binary signal; the rising edge of tooth $k = 0, \ldots, 57$ occurs at $\theta = 6k°$ (modulo 360°), the reference mark is the first rising edge after the gap.

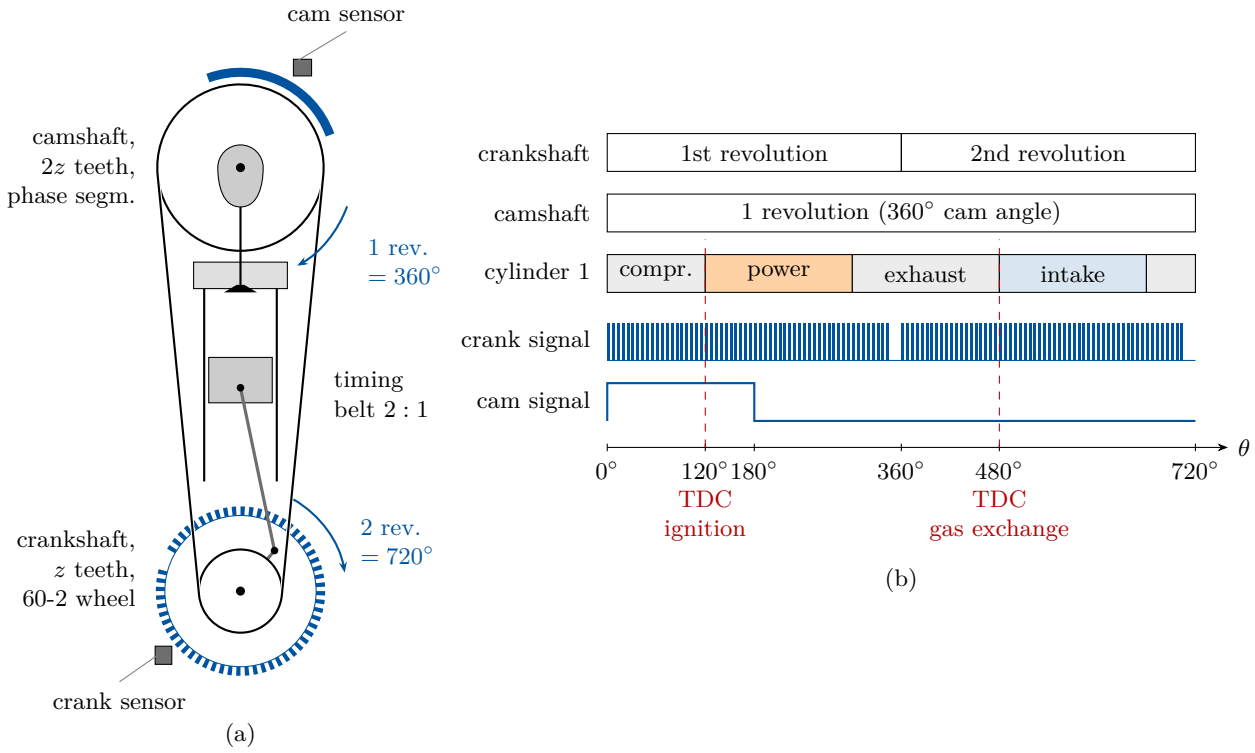


The tooth period T_k is measured from rising edge to rising edge. Because one tooth corresponds to $6° = 1/60$ revolution, the engine speed in rpm is

$$n = \frac{60\,\text{s/min}}{60\,T_k} = \frac{1}{T_k/\text{s}}\ \text{rpm}, \qquad n = \frac{3}{T_{\text{gap}}/\text{s}}\ \text{rpm (gap)}, \qquad n = \frac{60}{T_{\text{rev}}/\text{s}}\ \text{rpm (revolution)}. \tag{1}$$

Background: crankshaft, camshaft and the 720° cycle

A four-stroke engine needs *two* crankshaft revolutions for one working cycle (intake, compression, power, exhaust), but every valve opens only once per cycle. The camshaft is therefore driven by a timing belt (or chain) with a ratio of 2 : 1: its sprocket has twice as many teeth as the crankshaft sprocket, so it turns once while the crankshaft turns twice. The crank angle θ is counted over the full cycle, $0° \le \theta < 720°$, which is exactly one camshaft revolution (360° cam angle). The 60-2 wheel on the crankshaft repeats after 360° crank angle, the single segment of the cam phase wheel only after 720°.



(a) Schematic front view of one cylinder with crank drive and overhead camshaft; the timing belt turns the camshaft once per two crankshaft revolutions. (b) One working cycle of cylinder 1 over the crank angle: the crank wheel pattern and the piston position repeat after 360°, the strokes and the cam signal only after 720°.

Exercise 1.1 Signal classification and time scales

PAPER

a) Classify the five engine signals of Table 1.1 of the script (crank, cam, knock, ion current, lambda) with respect to (i) time: continuous/discrete, (ii) amplitude: analog/binary (digital), (iii) deterministic/stochastic, (iv) periodic/quasi-periodic/aperiodic. Collect the result in a table and justify every entry in one line. Note that the answer may differ for the physical sensor output and for the representation inside the ECU.

b) Discuss the borderline cases: (i) Is the crank signal periodic during an acceleration? Is it periodic in another independent variable? (ii) The knock signal is often called *quasi-periodic*, although a single knock event is a transient. Why? Introduce the notion of a signal whose *statistics* are periodic in the crank angle.

c) Fill in the following table for an engine speed of 3000 rpm: crank revolution period, tooth period and tooth frequency, duration of the gap (rising to rising edge), cam period, cam HIGH time, interval between two combustions (firing frequency), duration of 1° crank angle, time window of a knock onset (5°–25° after TDC), number of 6.5 kHz knock oscillations within one decay time $\tau = 0.8$ ms.

d) Crank and cam: the crankshaft turns 720° per engine cycle, the camshaft 360°. Why is the crank signal alone not sufficient to decide whether the piston of cylinder 1 is in its compression or in its exhaust stroke? After switching on the ignition with the engine cranking at 200 rpm, what is the worst-case time until the ECU knows (i) the crank angle modulo 360°, (ii) the full 720° phase, if the cam level is evaluated at $\theta = 60°$ after the reference mark?

Exercise 1.2 60-2 crank wheel and engine speed

WOKWI

Open the Wokwi project `code/wokwi/sheet01_crank/` (Arduino Uno, engine simulator chip, logic

analyzer, two LEDs, push button). `eng:CRANK` is connected to pin 2 (external interrupt INT0), `eng:CAM` to pin 4. The ISR `crank_isr()` is called at every rising crank edge and time-stamps it. With `TIMEBASE_TIMER1 = 1` the time stamp is the 16-bit Timer1 counter (prescaler 8, 0.5 µs per tick), with 0 it is `micros()` (resolution 4 µs on the Uno).

a) *Preparation (paper).* Derive (1). A timer of resolution Δ quantises the tooth period. Show that the resulting quantisation step of the per-tooth speed is $\Delta n \approx n^2\Delta$ (with n in rpm and Δ in s) and evaluate it for $\Delta = 4\,\mu s$ and $0.5\,\mu s$ at 600, 3000 and 7000 rpm. Compare with the amplitude of the 1 % speed ripple of the model. The gap is detected by the ratio test $T_k > 2\,T_{k-1}$: estimate the angular acceleration (in rpm/s) at which the test would fail (missed gap, or false gap), and compare with realistic engine accelerations ($\lesssim 10\,000$ rpm/s).

b) Complete the ISR: gap detection, tooth counter, per-revolution statistics, TDC LED (pin 12) on for 30° starting at each TDC edge. Which tooth numbers belong to $\theta = 120°$ and 300°? Check with the logic analyzer (D0 = crank, D2 = TDC output of the chip, D3 = your LED) and report the column `teeth` and `sync_err`.

c) Complete the speed computation in `loop()` (per revolution, last tooth, min/max per tooth). At 3000 rpm, compare `rpm_rev` with the spread `rpm_min`…`rpm_max` for both time bases. Press the button to dump 120 tooth periods, copy only this block and plot the per-tooth speed (column `rpm`), e.g. in a spreadsheet or with your CSV plot program of Bonus Exercise 0.B1 d) in JSLinux. Which part of the "jitter" is information, which part is measurement error?

d) Move the speed slider (600…7000 rpm) and repeat c) at 7000 rpm. Why is printing one CSV line per *tooth* not an option (estimate the UART load)?

Exercise 1.3 Cam/crank synchronisation CHALLENGE

Extend your sketch such that it determines the 720° phase from the cam signal (`eng:CAM`, HIGH for $0° \le \theta < 180°$).

a) Why should the cam level not be read at the reference edge ($\theta = 0°$), and which tooth is a good choice instead?

b) Implement `phase` (0° or 360°), toggle it at every gap, and check it at every revolution (`cam_err`). Light the LED on pin 11 only at the compression TDC of cylinder 1 and verify it with the logic analyzer (D1 = cam, D4 = LED).

c) Give the crank angle and the number of the cylinder that fires at each of the four TDC events.

Deliverables

1.1: completed classification table and time-scale table. 1.2: sketch, screenshot of the logic analyzer (gap, TDC, LED), table of rpm statistics for both time bases at 3000 and 7000 rpm, plot of a tooth-period dump. 1.3 (optional): extended sketch, logic analyzer screenshot with cam and cylinder-1 LED. Bonus 1.B1 (optional): program, table of statistics and periods, short interpretation.

Bonus: JSLinux (optional)

For interested students: the following exercise analyses all engine signals offline in JSLinux (set up as in Bonus Exercise 0.B1).

Exercise 1.B1 Signal zoo JSLINUX

The program `code/bonus/jslinux/sheet01/signal_zoo.c` generates 160 ms (four engine cycles) of the crank, cam, knock and ion signals at 3000 rpm, $f_s = 25$ kHz, and 20 s of a lambda sensor signal ($f_s = 200$ Hz) for a slowly oscillating $\lambda(t)$. Additionally, a crank signal during a speed ramp 2000 rpm to 4000 rpm is generated.

a) Complete `stats()`: mean, RMS value, minimum and maximum. Run the program and explain the values of the crank and cam signals analytically. Why is the mean of the knock signal not zero, although the knock sensor is AC coupled?

b) The normalised autocorrelation

$$r[m] = \frac{\hat{R}[m]}{\hat{R}[0]}, \qquad \hat{R}[m] = \frac{1}{N-m} \sum_{n=0}^{N-1-m} (x[n] - \bar{x})(x[n+m] - \bar{x})$$

is close to 1 at lags m where the signal resembles a shifted copy of itself. Complete `autocorr()` and `period_lag()` (first local maximum after the first zero crossing that reaches at least 90 % of the largest value). Report the estimated period and r for all six signals.

c) Interpret the results: Which signals are periodic, which only quasi-periodic? Why does the crank signal show a strong peak already at half a revolution? What happens to the periodicity of the crank signal during the ramp – and which lag is still detected? Why would the $1/N$-normalised (biased) estimate be misleading here?